

Name:	Maria [REDACTED]
Instructor:	Samuel Chukwuemeka
Objective:	To convert a measurement from a unit to another unit.
Measurement:	Volume
1st: Given Unit:	Customary unit (Fluid Ounce)
To Convert to:	Metric unit (Milliliters)
2nd: Given Unit:	Metric unit (Milliliters)
To Convert to:	Customary unit (Fluid Ounce)
Container Used:	Soft-soap body soap bottle (Please see bottom.)



(1.) Conversion: 20 fl. oz into mL

From the Given Tables:

$$1qt = 32 fl. oz$$

$$1gal = 4qt$$

$$1L = 0.26417205 gal$$

Method: Proportional Reasoning

First, I will convert Customary fluid ounces to Metric Liters.

These are the variables I will use.

x is the quantity of 20 fluid ounces in quarts.

a is the quantity of 0.625 quart in gallon.

b is the quantity of 0.15625 gallon in liter.

c is the quantity of 0.5914705965 liter in milliliter.

$$\frac{x}{20 fl. oz} = \frac{1qt}{32fl. oz}$$

Multiply both sides by 20 $fl. oz$ to solve for x .

$$20fl. oz \times \frac{x}{20 fl. oz} = \frac{1qt}{32fl. oz} \times 20fl. oz$$

$$x = \frac{1qt}{32fl. oz} \times 20fl. oz$$

$$x = 0.625qt$$

Using another variable on the next step to represent the solution.

$$\frac{a}{0.625qt} = \frac{1gal}{4qt}$$

Multiply both sides by $0.625qt$ to solve for a .

$$0.625qt \times \frac{a}{0.625qt} = \frac{1gal}{4qt} \times 0.625qt$$

$$a = \frac{1gal}{4qt} \times 0.625qt$$

$$a = 0.15625gal$$

Using another variable on the next step to represent the solution.

$$\frac{b}{0.15625gal} = \frac{1L}{0.26417205gal}$$

Multiply both sides by $0.15625gal$ to solve for b .

$$0.15625gal \times \frac{b}{0.15625gal} = \frac{1L}{0.26417205gal} \times 0.15625gal$$

$$b = \frac{1L}{0.26417205gal} \times 0.15625gal$$

$$b = 0.5914705965 L$$

From the Given Table:

$$1mL = 10^{-3}L$$

Conversion from Liters to Milliliters using the proportional method.

Using another variable on the next step to represent the solution.

$$\frac{c}{0.5914705965L} = \frac{1mL}{10^{-3}L}$$

Multiply both sides by $0.5914705965L$ to solve for c .

$$0.5914705965L \times \frac{c}{0.5914705965L} = \frac{1mL}{10^{-3}L} \times 0.5914705965L$$

$$c = \frac{1mL}{10^{-3}L} \times 0.5914705965L$$

$$c = 591.4705965 \text{ mL}$$

Rounding to the nearest whole number:

$$\approx 591 \text{ mL}$$

This confirms the quantity of fluid ounces to milliliters from the body soft-soap container.

(2.) Conversion: 591 mL into fl. oz

In the second conversation, I will convert Customary Metric Liters to fluid ounces.

These are the variables I will use.

a is the quantity of 591 milliliters to liters.

b is the quantity of 0.591 liters to gallons.

c is the quantity of 0.5914705965 gallons to quarter of a gallon.

d is the quantity of 0.6245027262 quarter of a gallon to fluid ounces.

From the Given Table:

$$1mL = 10^{-3}L$$

Method: Proportional Reasoning

First will convert Metric milliliters to liters.

$$\frac{a}{591mL} = \frac{10^{-3}L}{1mL}$$

Multiply both sides by $591mL$ to solve for a .

$$591mL \times \frac{a}{591mL} = \frac{10^{-3}L}{1mL} \times 591mL$$

$$a = \frac{10^{-3}L}{1mL} \times 591mL$$

$$a = 0.591L$$

Second will convert Metric Liters to Customary fluid ounces.

From the Given Tables:

$$1qt = 32 fl.oz$$

$$1gal = 4qt$$

$$1L = 0.26417205 gal$$

Using another variable on the next step to represent the solution.

$$\frac{b}{0.591L} = \frac{0.26417205gal}{1L}$$

Multiply both sides by $0.591L$ to solve for b .

$$0.591L \times \frac{b}{0.591L} = \frac{0.26417205gal}{1L} \times 0.591L$$

$$b = \frac{0.26417205gal}{1L} \times 0.591L$$

$$b = 0.1561256816gal$$

Using another variable on the next step to represent the solution.

$$\frac{c}{0.1561256816gal} = \frac{4qt}{1gal}$$

Multiply both sides by $0.1561256816gal$ to solve for c .

$$0.1561256816gal \times \frac{c}{0.1561256816gal} = \frac{4qt}{1gal} \times 0.1561256816gal$$

$$c = \frac{4qt}{1gal} \times 0.1561256816gal$$

$$c = 0.6245027262qt$$

Using another variable on the next step to represent the solution.

$$\frac{d}{0.6245027262qt} = \frac{32 fl. oz}{1qt}$$

Multiply both sides by $0.6245027262qt$ to solve for d .

$$0.6245027262qt \times \frac{d}{0.6245027262qt} = \frac{32 fl. oz}{1qt} \times 0.6245027262qt$$

$$d = \frac{32 fl. oz}{1qt} \times 0.6245027262qt$$

$$d = 19.98408724fl. oz$$

Rounding to the nearest whole number:

$$\approx 20 \text{ fl. oz}$$

This confirms the quantity of milliliters to fluid ounces from the body soft-soap container.